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Milestone 1: Computer Vision-Based Analysis of Volleyball Serve Trajectories

Agenda

- 1) Overview
- 2) Literature review
- 3) Dataset collection
- 4) Annotation scheme
- 5) Planned models
- 6) Next Steps

Overview

Goal: Build a vision tool for volleyball serves with a fixed camera

- Detect and track the ball across frames
- Segment court lines and map boundaries
- Reconstruct serve trajectory
- Landing frame and zone
- Evaluate consistency across serves



Literature Review

PathFinder / PathFinderPlus (H. Xia et al., 2023)

- Detects and classifies sets (setup for attack)
- Focused on tactics, not player analysis

TrackNet (Huang et al., 2019)

- Ball trajectory tracking via deep learning
- Handles tiny, fast objects well
- Multi-camera, heavy setup

Trajectory-Based Volleyball Tracking (B. Chakraborty and S. Meher, 2012)

- Detects ball trajectories
- Adaptable to serve trajectory analysis

BallTime AI

- Commercial ball tracking system
- Details + models not public

Dataset Collection

Collection Setup:

- Single camera in the center of the court
- Approx. 5m behind the court
- Captures entire court in frame

Pipeline

- Record raw sessions
- Split into individual serves
- Extract frames for annotating

Current Progress

- Approx. 30 serves between 3 players
- Started annotation work

Goal

- 200 serves across many players

Annotation Scheme

Ball

- Bounding box each frame

Landing

- Mark frame when ball hits the court

Court

- 6 corner points per session
- Used to map pixels to meters





Planned Models

Ball Detection & Tracking

- YOLOv8 to identify ball
- Tracks ball across toss → flight → landing

Calibration

- Homography using 6 annotated points
- Map pixel positions to real-world coordinates (meters)

Outputs

- Ball trajectories in real court space
- Predicted landing location
- Heatmap

Future: Automatic Court Segmentation

- U-Net for binary line masks
- Separates volleyball lines from background

Next Steps

Dataset

- Continue expanding dataset

Models

- Train YOLOv8 baseline for ball detection
- Use calibration to determine landing coordinates

Analysis

- Evaluate detection + tracking accuracy
- Generate serve trajectories + landing zones
- Measure consistency across players in sessions



Questions

